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中華電信股份有限公司
Chunghwa Telecom Co., Ltd.

Chunghwa Telecom Builds Multiple Generative AI Solutions with Amazon Bedrock

Chunghwa Telecom switched to Amazon Bedrock, accelerating the development of generative AI applications. These new applications include generating specifications documents for the software development lifecycle (SDLC), creating an interactive virtual English teacher, and developing a generative AI marketing assistant.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

13%

Less time spent on generating SDLC doc



Hi, I can connect you with an AWS representative or answer questions you have on AWS. ×

Higher accuracy

Improves consistency and quality of SDI



Need more info? Highlight any text to get an explanation generated with AWS generative AI. ×



Improved realism

Facilitates creation of more human-like virtual beings

Faster innovation

Provides base models to speed up development

Personalized marketing Overview

Automatically generates tailored marketing campaigns

[Chunghwa Telecom](#) is one of the largest integrated telco providers in Taiwan. To improve artificial intelligence (AI) data security and governance, the company migrated its generative AI projects to AWS. Chunghwa Telecom uses [Amazon Bedrock](#) for a variety of generative AI applications, including automatically generating specifications documents for the software development lifecycle and crafting custom marketing campaigns. With Amazon Bedrock, Chunghwa Telecom is saving developer hours and has also developed an immersive, interactive virtual English teacher for the first time.

Chunghwa Telecom uses Amazon Bedrock to build generative-AI applications

Opportunity | Seeking a Cloud Provider with Robust Data Security

As one of the largest integrated telco providers in Taiwan, Chunghwa Telecom provides not only consumer services like mobile and internet, but also other IT services such as cloud computing for enterprises.



Chunghwa Telecom has long been a strong supporter of industrial digital transformation and has been actively investing in generative artificial intelligence (AI) and large language models to enhance productivity and develop new products. The company established a cross-departmental team to research innovative applications for generative AI content and has fielded dozens of ideas from its employees for generative AI projects. However, its management determined that training a language model from scratch would take far too long for the competitive and fast-moving telco market.

Furthermore, Chunghwa Telecom needed a platform with robust security controls. Data protection and governance are of utmost concern for Chunghwa Telecom because many of its customers are government entities and financial firms.

Solution | Optimizing Generative AI Applications with Comprehensive Support

Chunghwa Telecom has been working with Amazon Web Services (AWS) for several years and chose [Amazon Bedrock](#) to build and scale new generative AI solutions. Jason Lin, vice president of information technology group at Chunghwa Telecom, explains, “The scope and scale of our engagement with AWS has increased, mainly because AWS has an incredibly secure cloud infrastructure, enhancing the speed and agility of security automation.”

Amazon Bedrock offers several capabilities to support Chunghwa Telecom’s security and privacy requirements. The service has achieved Health Insurance Portability and Accountability Act (HIPAA) eligibility and General Data Protection Regulation (GDPR) compliance—both important for the telco’s international partnerships. Additionally, Chunghwa Telecom values how Amazon Bedrock neither monitors or reviews content, nor uses it to improve the base models, nor shares content with third-party model providers.

Chunghwa Telecom also appreciates the guidance its teams have received from AWS Solutions Architects, [AWS Professional Services](#), [generative AI Innovation Center](#), AWS Worldwide Specialist Organization (WWSO), and industry solutions architects. The telco’s employees partook in several workshops, often in one-on-one settings, on application development and how to best use AI services and features within AWS.



Outcome | Designing Fully Immersive, Time-Saving Solutions

Since transitioning to Amazon Bedrock, Chunghwa Telecom innovated three generative AI applications: a software development lifecycle (SDLC) assistant, a virtual English teacher, and a generative AI marketing assistant.

The first generative AI solution, the SDLC assistant, has saved developers an estimated 13 percent of their time in generating SDLC documentation. Because Chunghwa Telecom has several hundred information systems, with several hundred projects completed each year, its developers typically juggle multiple assignments. Developers often logged overtime to write mandated SDLC documents, and the resulting fatigue contributed to manual errors.

Now, the SDLC assistant accelerates the speed of generating documents, improves document quality, and ensures consistency. “Compared to software development tools on the market, the assistant we’ve developed can better protect trade secrets, reduce costs, and meet the real-time needs of the business,” notes Lin. “We’re able to consistently add custom functions based on our internal regulations, to assist in interpreting software programs’ weaknesses, scanning for corrections, automatically reviewing APIs, and more.”

The second solution, the virtual English teacher, was designed as a unique, fully immersive experience. Chunghwa Telecom had prior experience creating virtual spokespersons, having deployed [Amazon Transcribe](#) and [Amazon Polly](#) to build speech-activated applications for customer service. For the “brain” of the virtual English teacher, the telco used [Amazon Titan](#) foundation models within Amazon Bedrock, along with [Stable Diffusion XL](#) text-to-image models, to create a more immersive experience. “Until recently, the most critical aspect behind authenticating virtual beings—the brain—always lacked a suitable solution,” Lin says.

“Since the introduction of Amazon Bedrock, we’ve been able to set personality traits for 2D/3D virtual characters, provide conversational knowledge, and animate interactive non-player characters (NPCs) that can be published across different platforms,” adds Lin. Chunghwa Telecom is working to apply these upgraded NPCs in scenarios such as retail interactions, product sales presentations, and corporate AI spokespersons.

Finally, the generative AI marketing assistant helps Chunghwa Telecom to craft unique marketing campaigns for each of its products. After using [Amazon Personalize](#) to help identify customer preferences, the assistant generates multi-dimensional tags which are categorized into personalized tags and product tags. Then, it utilizes generative AI to create personalized customer experiences without extra marketing costs—while also boosting click-through rates. “The assistant can quickly generate a variety of proposals—such as customized campaigns for SMS messages—expediting our ability to develop tailored campaigns early on,” Lin says.

Looking to the future, Chunghwa Telecom plans to continue cooperating with AWS as it establishes a competitive edge in generative AI and brings its technology into new industries and markets. Lin comments, “AWS provides not only powerful and flexible technical solutions but also deep support and professional advice, propelling us toward achieving our business goals and innovating more quickly. We highly value our relationship with AWS and look forward to achieving more success and breakthroughs soon.”

About Chunghwa Telecom

[Chunghwa](#) Telecom is one of the largest integrated telecommunications providers in Taiwan. With its expanding capabilities and modern technology offerings, the telco aims to create an optimal communication environment and facilitate seamless digital lifestyles for its customers.



AWS Services Used

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Stability AI, and Amazon via a single API.

[Learn more »](#)

AWS Professional Services

AWS Professional Service provides assistance through a collection of offerings which help you achieve specific outcomes related to enterprise cloud adoption.

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Amazon Personalize

You can now use Amazon Personalize and generative AI powered capabilities to quickly create hyper-personalization experiences for your users.

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Amazon Titan

The Amazon Titan family of models incorporates Amazon’s 25 years of experience innovating with AI and machine learning across its business.

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